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Summary

DevOps / Platform Engineer with production experience at Unstop, launched a user-facing cloud-native coding workspace platform. Recommended by Sr. Director and Sr. Engineers at Unstop. Building multi-region hybrid homelab infrastructure since high-school.

Experience

DevOps Intern | Unstop, Delhi

March 2026 - June 2026

- Built an AWS EKS platform that provisioned isolated coding environments for live assessments, supporting contest reliability with zero-downtime execution during active sessions.
- Built a Python (FastAPI) microservice to automate workspace lifecycle on AWS from startup to teardown, replacing stateless legacy orchestrator and reducing operational toil.
- Reduced workspace storage costs by ~ 80% using container images and identified an additional ~ 30-40% in AWS savings.
- Managed deployment & AWS infrastructure as code across EKS, VPC, EC2, and ECR using Terraform and Helm, with Karpenter handling automatic node scaling.
- Configured SigNoz (OpenTelemetry-based, Prometheus-compatible monitoring) for microservice observability, and built log pre-processing plus dashboards for faster troubleshooting.

Projects

Hybrid Infrastructure Lab

github.com/hudater/homelab/

- Operates hybrid infrastructure across on-premise hardware and Oracle Cloud, managing workloads as code with OpenTofu (Terraform), Terragrunt, and HashiCorp Cloud remote state.
- Connects on-premise and cloud sites via NetBird mesh, with Authentik enforcing authentication across all services; fully automated via OpenTofu (Terraform).
- Provisions and manages on-premise Kubernetes (k3s) clusters using OpenTofu, Ansible, and Bash for node configuration and lifecycle management.
- Self-hosts an AI inference stack on dedicated GPU hardware, using LLMProxy for unified routing between local (Ollama) and cloud LLM providers with centralized MCP configuration via MCPHub.

Recursive DNS & Observability Stack

blog.hudater.dev/posts/dns-infra/

- Operates a highly available recursive DNS cluster across both sites using Technitium, with state syncing over NetBird mesh.
- Runs Prometheus, Grafana, Loki, and Alloy for unified metrics, logs, and traces across homelab workloads to support monitoring and troubleshooting.
- Manages external & internal DNS for all domains (including portfolio at hudater.dev) on Cloudflare using OpenTofu and GitHub Actions CI/CD, with remote state in Hashicorp Cloud.

Open Source, Community & Publications

- **Linkme Generator** | github.com/ironicbadger/linkme/pulls?q=author:hudater
Contributed Golang (Go) features and GitHub Actions CI/CD pipeline to Linkme, an open source project by IronicBadger.
- **CNCG Noida SRE Games** | showcase.hudater.dev/sre_games.pdf
Diagnosed and resolved simulated Kubernetes microservice outages under time constraints using distributed tracing at CNCG Noida SRE Games using Dynatrace, Dash0 & Randoli.
- **Technical Writing** | blog.hudater.dev
Published technical articles on DNS internals, core networking protocols, and Git internals at personal blog.

Skills

- **Cloud & Infrastructure:** AWS, Oracle Cloud Infrastructure (OCI), Proxmox, Linux, High Availability, Cloud-Native Infrastructure
- **Containers & Orchestration:** Kubernetes, Docker, Helm, Karpenter
- **IaC, CI/CD & Version Control:** Terraform (OpenTofu), Terragrunt, Ansible, Git, GitHub Actions
- **Languages & Frameworks:** Go, Python, FastAPI, Bash
- **Networking & Security:** OPNsense, NetBird, WireGuard, VPC, VLAN, Traefik, Keycloak, Authentik
- **Observability:** SigNoz, Prometheus, Grafana, Loki, Tempo, Goldilocks (VPA)

Education

B. Tech CSE (Data Science) | Noida International University
8/10 GPA

Aug 2024 - Aug 2028